

On Handcrafted Machine Learning Features for Art Authentication

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Abstract—Accurate art authentication is crucial for ensuring correct attribution, cultural preservation, and financial valuation of artworks. Machine learning tools can significantly complement established techniques such as connoisseurship, provenance research, and scientific analysis (e.g., pigment analysis and radiographic imaging). This study explores a more objective and explainable approach to machine learning in art authentication. Specifically, we investigate the use of handcrafted features, such as image entropy, energy, texture patterns, and image similarities, to train a OneClassSVM model based on known authentic images of paintings by a given artist. To demonstrate the versatility of our method, we present results for three selected artists: John Constable, Vincent van Gogh, and Jean-Michel Basquiat. We further show by way of an example how our method can be utilised to study a given painting in terms of its attribution to a given artist. These examples, along with the accuracy of our results, indicate the potential of the proposed method as a flexible and efficient tool for art authentication alongside more established methods.

Index Terms—Art Authentication, One Class SVM, Machine Learning, Digital Heritage

I. INTRODUCTION

Art recognition and authentication involve a combination of historical research, scientific analysis, and expert evaluation. Historical research mainly focuses on the documentation and provenance of the artwork, tracing its ownership history and any records that verify its authenticity. Scientific methods include pigment analysis, infrared reflectography, X-ray fluorescence, and carbon dating to analyse the materials and methods used, providing clues on the age and origin of the piece in question.

Therefore, more recently, often in the absence of clear signatures or provenance records, a more objective and scientific approach to art recognition has emerged, keeping pace with technological advancements. This includes imaging techniques such as infrared reflectography and X-Ray radiography, elemental and molecular analysis such as Raman spectroscopy, and more recently, the application of traditional machine learning techniques and modern deep learning using neural networks [1].

A. Anomaly Detection

Anomaly detection in machine learning, within the context of art authentication, can be viewed as identifying deviations or outliers from the expected patterns in a given art piece that may indicate forgeries or alterations. This technique can use various types of classification algorithms to analyse image specific features such as, texture, colour distribution, brushstroke patterns, and other stylistic elements to detect inconsistencies that could suggest the artwork is not authentic.

As far as anomaly detection is concerned, generally, artwork authentication is compared only against authentic samples that can be considered as *normal*, whereas copies or forgeries will come out to be as *anomalous* when tested against such authentic samples. According to [2], *anomaly* or *outlierness* may have several definitions, as it can be characterised by means of (a) its scarcity, as anomalous/novel/outlier events occur less frequently than “normal” events; (b) its characteristics, as anomalous/novel/outlier events should have different characteristics than “normal” events; (c) its meaning, as such events should carry a specific and a different meaning than normal events. However, in the context of artwork authentication, an *Anomalous/Outlier* sample can be defined as that whose features are too distant from those of *authentic/normal* samples. In this case, *anomaly detection* stands as a one-class classification, and the resulting machine learning model should be trained on *normal/authentic* data only. Several anomaly detection methods have been proposed in the literature, including one-class support vector machines (OC-SVM), random forests, and auto-regressive autoencoders. These methods are generally trained on a set of *normal* samples only.

In particular, OC-SVM has been successfully applied as an anomaly/outlier/novelty detection method for several image processing applications, such as face authentication [3], tumour detection in medical imaging [4], and more recently in active mobile phone user authentication [5].

B. Scope of the Paper

This study aims to advance the field of art authentication by introducing a machine learning approach emphasising the use of handcrafted feature selection. Image features such as colour variation, hues, tonal values, image entropy, energy, texture patterns, and image similarities could be crucial inputs for machine learning models. Therefore, this research explores the potential of such features to be incorporated into a carefully chosen machine learning framework to enhance the accuracy, reliability, and explainability of the algorithmic decision.

The rest of the paper is organised as follows. Section II reviews the related work. Section III presents the methodology, including the method utilised to obtain the selected features and how they are extracted. Then, Section IV details the experimental protocol and Section V presents the results obtained in this study. Finally, the work is further discussed and concluded in VI.

II. RELATED WORK

Machine learning techniques like convolution neural network based deep learning offer significant potential for art authentication through image analysis and pattern recognition [6], [7]. These models can analyse large datasets of various artists, detecting intricate patterns and features that may elude human experts. The study by Ugail et al., [1] primarily focuses on deep learning techniques and edge detection methods such as Canny edge, Laplacian, Scharr operator and Sobel operator and achieves a high accuracy 98% using ResNet50 in a binary classification task of authenticating paintings by Raphael. While this study does incorporate some handcrafted features at a lower level in the inference process, it primarily relies on them to support the underlying deep learning model. This reliance suggests a potential under-utilisation of handcrafted edge features, which can independently provide additional accuracy and possibly explainability.

Similarly, Dobbs et al., [8] trained and tested the ResNet101 model on 45,974 paintings from 90 artists, and it achieved an overall accuracy of 87.31%, outperforming the ResNet18 baseline model. The dataset used in this study is imbalanced, with some artists having significantly more artwork than others. This imbalance can clearly affect the performance of models. Challenges in obtaining sufficient high-quality training data may often prove difficult in the context of art analysis, where the number of available artworks might be limited. It is thought, in this context, too, that having reliable handcrafted features for machine learning may be an added advantage.

Handcrafted features are inherently interpretable, as they are based on well-understood statistical and geometric properties of the image features. This allows art historians and researchers to draw direct correlations between the features and specific artistic techniques, making the analysis more transparent. Handcrafted features can be tailored specifically to capture the digital signature and style of artistic brushstrokes and textures.

Computational techniques, such as graylevel co-occurrence matrix (GLCM) and wavelet transforms, are increasingly used to analyse texture and brushstroke patterns in paintings. Zhong et al., [9] proposed the use of a four-directional GLCM to detect brushstroke information, which is used in conjunction with an RGB channel to classify art in the domains of style and genre. Kim et al., [11] discuss the use of edge-based procedural textures by focusing on edges, which results in a 60% faster synthesis speed. Toennies [12] discusses the various texture analysis methods such as SIFT, Histogram of Gradients (HOG), Gabor, and Local Binary Pattern (LPB). In a study on the use of Haralick features of footprints, the use of Harilick features improves the Genuine Accept Rate (GAR) by 5.5% [13]. In this context, Mansour & Thomson [14] employed Haralick features to quantitatively analyse the patterns of specific energy characteristic of the radiation sources utilised in the study of the spatial distribution of dosimetric data. By leveraging these texture analysis features, they aimed to capture and measure the distinct energy patterns associated with the various radiation sources under investigation, aiding in deeper analysis of the dosimetric distributions across cellular to tumour length scales. Similarly, the study by Hapsari et al., [15] demonstrates that measures such as contrast, entropy, and homogeneity can quantify differences in specific energy distributions arising from various radiation

sources at the cellular scale. At the tumour scale, such features can quantify differences in absorbed dose distributions due to factors like tissue heterogeneities (e.g., calcification) and simulation conditions.

Similarly, features like intensity histograms and shape descriptors appear to enhance the classification accuracy for distinguishing between benign and malignant breast tumours. For example, the DenseXtNet model trained using texture features such as energy, entropy, correlation, homogeneity and inertia is reported to have achieved an accuracy of 98.6% in classification [16]. Other studies, using image entropy, spatial features (e.g., area and perimeter) and intensity features (e.g., the mean, standard deviation of fluorescence) show the potential use of handcrafted features for task-oriented image analysis, [17], [18] and [19].

In the context of image analysis, experimental results on several benchmark texture datasets (e.g., Outex, Brodatz, and VisTex) showed that incorporating GLCM features as part of the regional input features improved the texture recognition accuracy compared to using only LBP or Gabor features [20]. This approach achieved state-of-the-art performance on the Outex and Brodatz datasets, outperforming several existing texture recognition methods. Similarly, In their study, Asmus & Parfenov [21], [22] propose the use of intensity histograms from digital images of paintings, which serve as unique “fingerprints” of an artist’s technique.

III. METHODOLOGY

In this study, we employ the One-Class Support Vector Machines (OC-SVM) to develop anomaly detection models aimed at understanding the authenticity of a given painting. Given a dataset of images corresponding to authentic paintings of an artist, a distinct OC-SVM model is trained. The model training process utilises several handcrafted features that are carefully chosen and thought to capture the most distinguishing image characteristics of the paintings by the artist.

A. One-class SVM

The OC-SVM is a variation of SVM methods called one-class support vector machines that seeks to estimate a function with positive values on one half-space and negative values on its complement. OC-SVM typically separates the input space into two categories: *outliers* and *normal* data. The main difference to regular (or binary/multi-class) SVM consists of training OC-SVM on *normal* data only. Then, the decision on the degree of “outlierness” is made upon the sign of the function $g(x)$, such that,

$$g(x) = \text{sign}(w^T \phi(x) - \rho), \quad (1)$$

where ϕ is the kernel function, w is the orthogonal vector to the separating hyperplane, and ρ is a bias term. An input sample x is classified as *normal* if this function is positive; otherwise, it is an outlier [23].

B. Feature Selection

Following a preliminary study of a number of image features, we decided to utilise three types of features: a) texture features, b) a similarity feature and c) the Gray-Level Co-occurrence Matrix (GLCM). Specific texture features such as image entropy, contrast, and correlation were found to be

particularly informative for distinguishing graph classes and characterising network [24].

Therefore, four handcrafted features were selected, namely the entropy, energy, contrast and homogeneity. Given an image $P(i, j)$ with dimensions $M \times N$, the following metrics are used to compute these features.

Fourier Energy: Given the 2D Fourier Transform of the image $P(i, j)$ is described as,

$$F(u, v) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} P(i, j) e^{-2\pi i \left(\frac{ui}{M} + \frac{vj}{N} \right)}, \quad (2)$$

and the magnitude of the Fourier Transform is given as,

$$|F(u, v)| = \sqrt{\Re(F(u, v))^2 + \Im(F(u, v))^2}, \quad (3)$$

then, the Fourier energy of $P(i, j)$ is,

$$E = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} (\Re(F(u, v))^2 + \Im(F(u, v))^2). \quad (4)$$

Entropy: Given the probability distribution of the pixel intensities is described as,

$$p(k) = \frac{h(k)}{M \cdot N}, \quad (5)$$

then the entropy H of the image is,

$$H = - \sum_{k=0}^{L-1} \left(\frac{h(k)}{M \cdot N} \right) \log_2 \left(\frac{h(k)}{M \cdot N} \right). \quad (6)$$

Contrast: If we consider the mean intensity of $P(i, j)$ such that,

$$\mu = \frac{1}{M \cdot N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} P(i, j), \quad (7)$$

then, the standard deviation of the pixel intensities is,

$$\sigma = \sqrt{\frac{1}{M \cdot N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (P(i, j) - \mu)^2}. \quad (8)$$

Homogeneity: Assuming the normalized Gray Level Co-occurrence Matrix (GLCM) is given by,

$$P(i, j) = \frac{C(i, j)}{\sum_{i=0}^{L-1} \sum_{j=0}^{L-1} C(i, j)}, \quad (9)$$

then, the homogeneity of the image is,

$$H = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} \frac{P(i, j)}{1 + |i - j|}. \quad (10)$$

Similarity Feature: In addition to the above image features, we also enhanced the feature set by introducing the cosine similarity measure of ResNet50 features [25]. This is done by selecting n similar images in the training image for a given image.

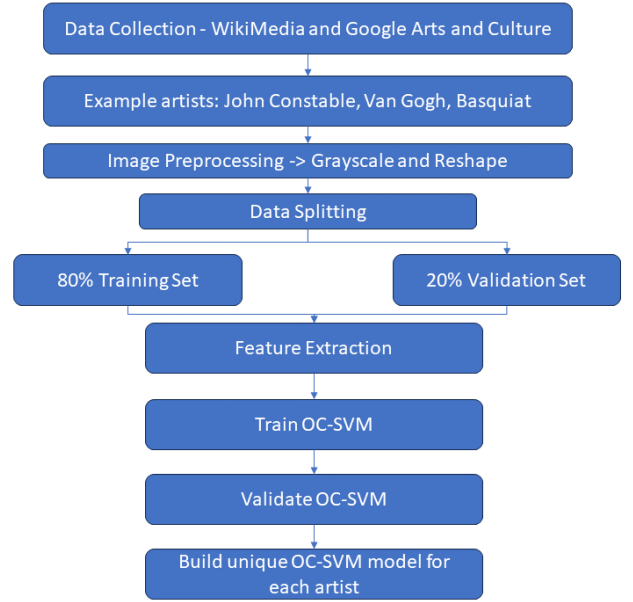


Fig. 1. A flowchart describing the processes involved in the OC-SVM classifier training, validation and inference.

IV. EXPERIMENTAL PROTOCOL

The process depicted in the flowchart, as shown in Fig. 1, outlines the chosen approach for creating and validating the OC-SVM models to analyse images of artworks from various artists. The workflow begins with data collection from reputable copyright-free sources such as Wikimedia and Google Arts and Culture. For example, in this work, we have mainly focused on three notable artists, i.e., John Constable, Vincent van Gogh, and Jean-Michel Basquiat. These examples are specifically chosen to showcase the ability of the OC-SVM models to cater for the range and breadth of artistic styles they have employed. Fig. 2 shows a small sample of data from the three artists. This collected data undergoes an initial image preprocessing phase, where the images are converted to grayscale and resized to a consistent format suitable for analysis. Once the preprocessing is complete, the dataset is split into two subsets: an 80% training set and a 20% validation set.

In the subsequent steps, feature extraction is performed. In Fig. 3, we show a sample feature set for a typical image in the dataset. Here, we have taken an image from the van Gogh image dataset, identified the three most similar images using the cosine similarity measure of the ResNet50 features, and then computed their mean energy features.

Once we compute the features, the OC-SVM will be trained and validated. The final step involves creating an artist specific model that can be used for inference purposes.

A. Data Preprocessing and Feature Selection

In the preprocessing phase, all images were initially converted to grayscale to simplify the computational process and emphasise the core features essential for analysis. Grayscale conversion reduces the complexity of the image data by eliminating colour information, which is often unnecessary for many image processing tasks. Thus, the focus is on intensity variations. This transformation helps highlight the structural and textural aspects of the images, which are crucial for further feature extraction and analysis.

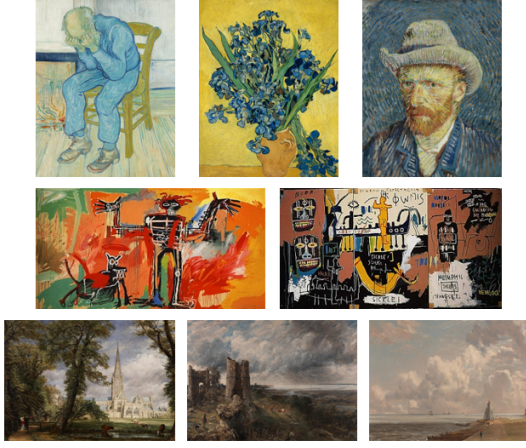


Fig. 2. A small sample from the datasets used for training the models. In this work, we have used artwork from three notable artists, i.e., John Constable, Vincent van Gogh, and Jean-Michel Basquiat. These examples are specifically chosen to showcase the ability of the OC-SVM models to cater for the range and breadth of artistic styles they have employed.

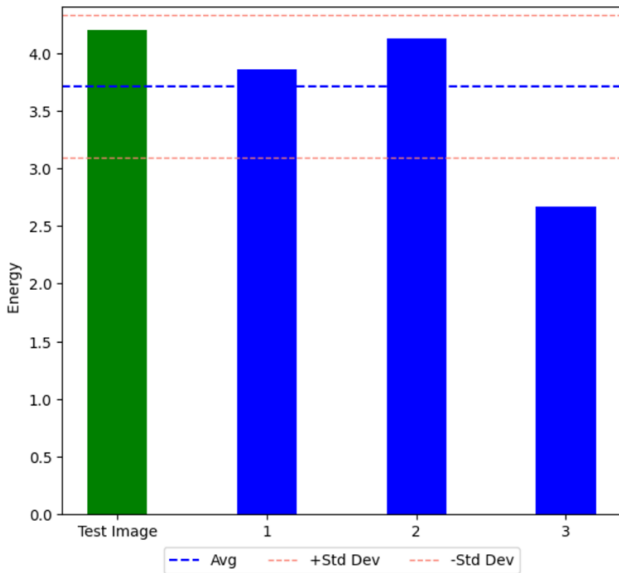


Fig. 3. A sample feature set for a typical image in the dataset. Here, we have taken an image from the van Gogh image dataset (Test Image), identified the three most similar images (1, 2, 3) using the cosine similarity measure of ResNet50 features, and then computed their energy features. The mean energy value, +1 and -1 standard deviation from the mean are also shown.

Table I shows the various types of features extracted from the images, including the entropy, energy, contrast, and homogeneity, each quantified with mean and maximum values. Additionally, features from the contours and similarity measures based on ResNet50 are included, culminating in a total of 14 distinct features. This “optimal” feature set is selected following extensive experimentation using various image analysis features and tools.

B. Classifier Training and Validation

Prior to applying features in the training routine, we normalised them using a fitted scaler to ensure consistency and improve the model’s performance. During the training phase, we conducted a grid search with cross-validation (CV=4) to optimise hyperparameters such as kernel type, gamma, and

nu, which are crucial for fine-tuning the OC-SVM model. This process enabled us to find the best combination of parameters that maximise the model’s accuracy and generalisation capabilities. Consequently, the OC-SVM was trained on features derived from reference images, which were treated as inliers, forming the basis for anomaly detection.

As previously discussed, the One-Class Support Vector Machine (OC-SVM) model we utilised was trained using the extracted features to identify anomalies directly within the feature space. This approach is effective in distinguishing between normal and anomalous data points based on the learned feature distribution. A similar technique was employed by Mabu et al. [26] for landslide area detection, demonstrating its versatility across different application domains.

Thus, to validate our models, we begin by processing all the images in the training set through OC-SVM pipeline. We compute the mean and standard deviations of the OC-SVM function output to establish a baseline for normal data. This initial evaluation allowed us to assess the behaviour of the models with known data and provided a reference for subsequent anomaly detection.

Note that the decision scores OC-SVM classifier indicate the distance of the feature vector from the learned decision boundary, with negative scores typically signalling an outlier or anomaly. Additionally, we compute the mean and standard deviation of a given score from the mean. The standard deviation from the mean enables us to establish a threshold to determine an acceptable range for inliers, classifying any score outside this boundary with respect to the selected standard deviation criteria as an outlier.

V. EXPERIMENTAL RESULTS

We used three distinct datasets comprising artworks by Constable, van Gogh, and Basquiat to train and evaluate our models. For each artist, an OC-SVM model was developed. The training phase involved normalising the features with a fitted scalar and performing a grid search with cross-validation to optimise key hyperparameters such as kernel type, gamma, and nu. Tables II, III and IV summarise the results obtained on the test images.

A. An Example Case Study

The example depicted in Fig. 4 shows how our method can be utilised to study a given painting in terms of its attribution to a given artist. This particular painting is known as The Last Farewell of Maria Constable and, to our knowledge, has not been subject to machine learning analysis previously, and it serves as a good example against which to run our method.

To begin the analysis, features were extracted from the painting using image processing techniques. The extracted features were then scaled and processed using a previously trained Constable OC-SVM model. The model then generated an inference, providing the prediction score.

Fig. 5 presents a histogram of the OC-SVM results for this case study example. The results show that the prediction score for this painting falls well within +1 standard deviation of the mean score for authentic Constable paintings, suggesting a high probability that the painting is genuine. This placement indicates that the painting’s features closely align with those of the authentic training set, affirming its likely attribution to John Constable.

Table I. Details of the features selected for training the classifier. These include image level features, contour level features and ResNet50 similarities.

Feature Type	Features	Count
Image	Entropy (mean and max)	2
	Energy (mean and max)	2
	Contrast (mean)	1
	Homogeneity (mean and max)	2
Contour	Entropy (mean and max)	2
	Energy (mean and max)	2
Similarity	ResNet50 similarity (for 3 most similar images)	3
Total features		14

Table II. Results of the analysis for the test images used in the Constable OC-SVM model.

Metric	Total	+1 Std Dev	+2 Std Dev	+3 Std Dev
Number of test images	48	42	45	46
Prevalence	100%	87.50%	93.75%	95.83%
Validation accuracy	95.60%	94.50%	95.70%	96.60%

Table III. Results of the analysis for the test images used in the Van Gogh OC-SVM model.

	Total	+1 Std Dev	+2 Std Dev	+3 Std Dev
Number of test images	49	42	44	49
Prevalence	100%	85.71%	89.80%	100%
Validation accuracy	96.40%	95.00%	96.00%	98.20%

Table IV. Results of the analysis for test images in the Basquiat OC-SVM model.

	Total	+1 Std Dev	+2 Std Dev	+3 Std Dev
Number of test images	71	58	66	69
Prevalence	100%	81.69%	92.96%	97.18%
Validation accuracy	95.8%	95.00%	96.10%	96.30%



Fig. 4. Image of a test painting thought to have been painted by John Constable. The painting is known as The Last Farewell of Maria Constable. This painting has not previously been subject to machine learning based authentication analysis. Image: courtesy of Anne and Wolfgang Voit.

VI. CONCLUSIONS

In this study, we have introduced a novel approach to painting authentication, leveraging anomaly detection through OC-SVM and a carefully selected set of machine learning-extracted features, including entropy, energy, contrast, and homogeneity. Our results indicate high accuracy and precision across various error thresholds, ranging from 1 to 3 standard deviations.

This work has demonstrated the effectiveness of OC-SVM models in art authentication by employing handcrafted

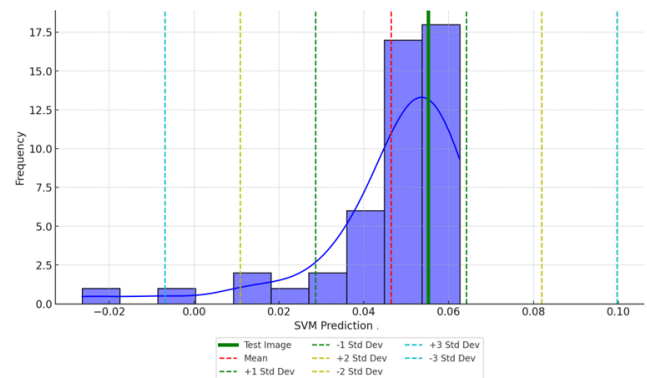


Fig. 5. Histogram showing the OC-SVM results for the image of the painting The Last Farewell of Maria Constable. We note the OC-SVM result for this painting falls well within +1 standard deviation of the mean, indicating a high probability of it being a genuine Constable.

features for anomaly detection. We have shown that our machine learning approach can effectively carry forth correct attribution by focusing on the works of renowned artists like John Constable, Vincent van Gogh, and Jean-Michel Basquiat.

The feature set chosen for this study resulted from extensive experimentation with various image analysis techniques. While promising, it is not definitive and requires further research and testing to validate and improve its effectiveness, particularly given the small number of analysed images. The influence of dataset size on overall accuracy has yet to be

fully understood; thus, ensuring a representative and diverse sample is crucial for dependable results in anomaly detection tasks.

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